SCI-CAL

Detector Calibration, Atmospheric Extinction, and Signal Transfer

Engineering Conformance

Version 0.1 - DRAFT

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An implementation-independent conformance view of the same shared scientific authority, suitable for a clean implementation or future audit without mapping to current files, functions, classes, or behavior.

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Independent, implementation-blind authorship

Draft authority boundary. This is an implementation-blind scientific contract draft. It defines proposed scientific meaning and falsifiable conformance conditions. It is not evidence of implementation conformity, atmosphere-model fidelity, observational repeatability, absolute flux accuracy, production readiness, or completed validation.

Abstract

This document converts the shared SCI-CAL v0.1 scientific authority into observable conformance conditions. A candidate realizes a typed, occurrence- scoped transformation from admitted `xs` samples to top-of-atmosphere point-source-peak mJy beam^{-1}, applies selected `flxscale` and target atmosphere correction exactly once, propagates conditional uncertainty with the identical operator, maintains a complete nuisance ledger, and emits resolvable lineage and truthful validity/quality state. This view does not prescribe present software structure and does not assert that any implementation conforms. The common numbered requirements are the only normative clauses; the procedures here describe evidence for them.

Contents

1 Conformance object and non-claims

The object under review is one complete SCI-CAL transformation for a coherent observation or declared segment, including its inputs, resolved state, realized numerical outputs, uncertainty products, validity/quality state, canonical lineage, and compact product links. A numeric array alone is not a conformance object.

Conformance is evaluated against the common definitions, equations, assumptions, requirements, and edge predictions included below. The review does not infer scientific meaning from names, defaults, row positions, file placement, or historical behavior. It does not map requirements to current files, functions, classes, configuration keys, or tests. Such a mapping, if later authorized, is a separate implementation audit.

A structural pass establishes only the first claim layer in Table 3. Atmosphere representation and observational performance require their own evidence. Its strongest structural-policy state is **science-qualification-eligible**, never an achieved `science-qualified` or `calibrated-science` claim. The latter claims remain governed by the separate-evidence threshold in `SCI-CAL-REQ-049`. No production state follows from this draft.

2 Logical conformance records

The following are logical records. They may be serialized or implemented in any form that preserves the stated semantics and relationships.

Table 1: Logical records to expose for conformance review.

Record	Minimum semantic contents	Primary checks
Requested state	requested channel, target unit/meaning, mode, source choices, and explicitly supplied expert values	immutable acceptance; unsupported requests remain visible
Effective state	resolved activation, legal unit path, normalized factor direction, selected operator policy, and rejection diagnostics	one-way resolution; “automatic” resolves to an exact effective choice
Observation-resolved state	observation/Tune, acquisition keys, column binding evidence, selected APT identity/digest, row occurrences, association edges, factors, atmosphere/time support, response basis, and uncertainty ledger	complete replacement per observation; unique keys and supported domains
Realized factor ledger	one entry per factor instance and occurrence/support, including role, value, unit, direction, reference plane, stage, parent lineage, validity, and application count	Equation (14); no embodied or parent double application
Realized output state	valid occurrence set, calibrated values, conditional uncertainty or unavailable state, segment quality, reason codes, and response status	Equations (13), (17), and (1)
Canonical lineage	all resolvable identities and relationships in Equation (29) plus nuisance completeness/correlation scope	deterministic reconstruction and no stale observation state

Record	Minimum semantic contents	Primary checks
Product link	canonical-record reference, selected APT reference, and occurrence keys sufficient to resolve both	resolvable without copying the entire per-detector table

3 Conformance evaluation sequence

The sequence below is explanatory rather than an implementation architecture. It groups the observables that the common requirements place at the cold boundary before a hot sample calculation can have scientific meaning.

E1. Normalize the request. Establish that the measured stream is `xs`, the target is top-of-atmosphere point-source-peak mJy beam^{-1}, and the requested/effective distinction is recorded.

E2. Resolve occurrence identity. Validate acquisition keys, the column relation, selected immutable APT identity, local row occurrences, target-to-source edges, and array/network constraints. Establish exact cardinality for the selected support before evaluating factors.

E3. Resolve factor causality. Admit one selected `flxscale` per detector occurrence, verify its direction and reference plane, classify the pointing-correction ancestry, and reject opaque or duplicate factor roles.

E4. Resolve atmosphere support. Validate opacity and airmass meaning, time coverage/bracketing, segment quality, exact content-bound operator, node support, ordinate orientation, and passband identity/limitations. In the present draft, `SCI-CAL-ASM-011` necessarily stops numeric calibration here.

E5. Evaluate once. On the admitted support, evaluate Equations (7), (12), and (13). Record factor-instance application counts and numeric factor ranges. Invalid occurrences remain outside the domain rather than becoming zeros.

E6. Propagate conditional uncertainty. Apply Equation (17) or (18) to the identical support. Preserve full correlations when supplied; otherwise record the exact available form or an unavailable state.

E7. Resolve nuisance and response claims. Complete the required nuisance ledger, retain correlations, link the originating Beammap response separately from any realized downstream response, and withhold total or point-source-peak claims that lack their conditions.

E8. Publish reconstructible state. Emit the validity tuple, reason codes, one canonical lineage record, and resolvable product links. State the status of structural, representation, and observational claims separately.

4 Notation and typed quantities

The contract uses occurrence-scoped identities. An index is never, by itself, a persistent physical identity. Let o denote an observation or Tune context, t a location on its approved common sample axis, d a selected detector occurrence, $a(d)$ its TolTEC array, and $j = (o, t, d)$ a compound sample occurrence. The three arrays are $a1100$, $a1400$, and $a2000$. Networks 0 through 6 belong to $a1100$, 7 through 10 to $a1400$, and 11 through 12 to $a2000$.

Table 2: Canonical symbols and types.

Symbol	Identity and support	Unit	Meaning
x_j^{xs}	acquisition sample occurrence	declared pre-CAL unit $U_{x,d}$	admitted ordinary `xs` detector sample
b_j	same recipient/support as x_j	$U_{x,d}$	explicitly authorized offset or baseline; “not applicable” is distinct from missing
$z_j = x_j^{xs} - b_j$	valid sample support	$U_{x,d}$	CAL input after the declared affine convention
K_{od}^{acq}	target acquisition occurrence	identity only	observation/Tune, network or interface, and network-local tone or row slot

Symbol	Identity and support	Unit	Meaning
R_{sd}^{APT}	selected source-row occurrence	identity only	immutable APT identity plus source observation/network context and artifact-local row key
$E_{od,sd}^{\mathrm{assoc}}$	target-to-source occurrence edge	identity only	matcher method/version, disposition, and available quality evidence
F_{od}^{sel}	selected APT row occurrence	$\mathrm{mJy\,beam}^{-1}/U_{x,d}$	authoritative absolute `flxscale` for this selected lineage
P_{od}	lineage event, if applicable	dimensionless	TolProj pointing-derived correction already embodied in F^{sel}; never an additional CAL runtime factor
r_{od}	selected APT row occurrence	declared relative-response unit	`responsivity`; a relative quantity, not absolute flux calibration
s_{od}	selected APT row occurrence	declared sensitivity unit	Beammap `sens`; sensitivity or approximate-weight input only
$\tau_{225,o}(t)$	declared time support	dimensionless	zenith optical depth at 225 GHz
X_{ot}	eligible sample occurrence	dimensionless	full sample airmass supplied under approved upstream meaning
$\ell_{ot} = \tau_{225,o}(t)X_{ot}$	eligible sample occurrence	dimensionless	line-of-sight 225 GHz optical-depth coordinate relative to $X_{\mathrm{ref}} = 0$
$\mathcal{O}_a$	content-bound operator for array a	identity plus nodes	retained `am12_fixed_djf25_piecewise_linear_los_tau_v1` structural operator
$H_a(\ell)$	operator node support	dimensionless	authoritative piecewise-linear ordinate, whose orientation must be declared as transmission or correction
$T_a(\ell)$	operator support	dimensionless	band transmission implied by H_a and its declared orientation
$C_a(\ell) = T_a(\ell)^{-1}$	operator support	dimensionless	positive correction from the eligible sample plane to top of atmosphere
M_j	valid sample occurrence	$\mathrm{mJy\,beam}^{-1}/U_{x,d}$	exact realized once-only multiplier
y_j	valid sample occurrence	$\mathrm{mJy\,beam}^{-1}$	calibrated top-of-atmosphere, point-source-peak detector sample
$C_{x\|\theta}$	valid input vector support	U_x^2 with typed blocks	conditional measurement covariance at fixed calibration state θ
$C_{y\|\theta}$	identical valid output support	$(\mathrm{mJy\,beam}^{-1})^2$	transformed conditional measurement covariance
$v_j^{(x)}, w_j^{(x)}$	scalar sample occurrence	U_x^2, U_x^{-2}	diagonal conditional variance and inverse variance, when supplied
θ	named detector, array, observation, cohort, or global scope	parameter-specific	calibration, atmosphere, and response nuisance state

Bold symbols stack only the declared valid occurrences; they do not insert numeric zeros for invalid samples. A locator such as a global column, table row, or map index may find an occurrence, but cannot establish its identity.

5 Authoritative definitions and claim boundaries

5.1 Operation and endpoint

SCI-CAL v0.1 operation. On admitted `xs` support, SCI-CAL binds each target acquisition occurrence to one row occurrence in one exact selected immutable measured APT, applies that row's absolute `flxscale` once, applies one supported target-observation atmosphere correction to the top-of-atmosphere plane, and propagates conditional measurement uncertainty. It ends at calibrated detector samples, not at a map, fitted source, or integrated photometric estimate.

Point-source-peak mJy beam^{-1}**.** The label denotes a peak-normalized response convention. For an unresolved source of top-of-atmosphere flux density S mJy, a downstream map whose realized beam, mapmaker, kernel, and filter response has unit peak for the declared response basis has ideal peak S mJy beam^{-1}. This definition does not authorize surface-brightness, per-pixel, temperature, extended-source, or integrated-flux interpretation.

Reference plane. The target is top of atmosphere with $X_{\mathrm{ref}} = 0$. A target sample at airmass X_{ot} is corrected from its observing plane to that reference. Atmosphere corrections used while deriving the selected source APT belong to that APT's ancestry; they do not replace or duplicate the target-observation correction.

5.2 Identity and immutable selection

Acquisition occurrence. The tuple of observation/Tune context, network or interface, and network-local channel/tone slot. It identifies the target measured occurrence to which a sample column is bound.

Measured-APT row occurrence. The tuple of immutable selected-artifact identity, its artifact-local row key, and source observation/network context. A UID is local to that immutable artifact. It is not a universal detector name.

Association edge. A separate, directed target-to-source relation that names both occurrence-scoped endpoints, matcher method and version, disposition, and available quality evidence. Only a unique admitted match is sufficient for calibration. Aggregate matcher performance is not a per-row match probability.

Identity layers. Acquisition identity, measured-APT identity, cross-observation association, and optional design identity answer different questions. Occurrence identity, order-independent semantic-content identity, and exact byte-transport identity are also distinct. Equal row numbers, local keys, paths, timestamps, or integer spellings do not prove correspondence.

Binding. Either an explicit keyed relation or a verified ordered-row contract may bind acquisition occurrences to sample columns and selected APT rows. An ordered relation is valid only with recorded scope, cardinality, ordering definition, and proof. Row position alone is not identity.

5.3 Factors and once-only lineage

Factor instance. A value together with a stable instance identity, scientific role, direction, unit, recipient identity, support, reference plane, validity, application stage, uncertainty status, and parent lineage. Numerically equal values with different roles are not interchangeable.

Selected absolute factor. F^{sel}_{od} is the selected APT row's `flxscale`, oriented as mJy beam$^{-1}/U_{x,d}$. It is the sole absolute detector factor in the v0.1 signal multiplier. If TolProj created a corrected child APT, its pointing-derived correction is an ancestry event already embodied in this value.

Relative responsivity. `responsivity` may support a separately declared relative-response, diagnostic, normalization, or nuisance role. It does not independently establish an absolute mJy beam^{-1} scale and is not an additional v0.1 signal multiplier.

Sensitivity. Beammap `sens` is a separately typed sensitivity or approximate-weight quantity. It is neither atmospheric extinction nor an absolute signal factor nor a general total uncertainty.

Opaque total factor. An unqualified `fcf` is not authoritative. A compatibility value can be admitted only when its exact contents, units, direction, exclusions, and constituent instance identities reproduce the canonical factorization without omission or duplication.

Once-only composition. Each required physical role contributes once to a realized multiplier. Causal derivation order is recorded in lineage; numeric multiplication of admitted scalar factors is order independent. A factor already materialized into a selected artifact is not multiplied again at execution.

5.4 Atmosphere operator and passband status

Structural operator. The retained operator identity `am12_fixed_djf25_piecewise_linear_los_tau_v1` is adopted only as a piecewise-linear function of line-of-sight optical depth with an analytic zero anchor, exact declared nodes, continuity, and positive finite transmission and correction. Its content-bound node record, ordinate orientation, and numeric support are not present in the author packet and therefore remain unresolved for operational evaluation in this draft.

Passband reference. The only selected passband identity is

`toltec-passband-set-v1:sha256:`
`5e6f38f14bcae93a29ffe8362c52b15209f51aee4e48373b23aaa5ec2f8a6433`.

It binds one array-named modeled curve for each TolTEC array and an index. The frequency coordinate is in GHz and the supplied throughput is dimensionless. The curves are neither peak-normalized nor area-normalized. Detector or network aggregation, telescope measurement, uncertainty and covariance, normalization semantics, original generation recipe, and photon-versus-energy physical convention are not established. The passband identity can identify an input to a declared operator; it cannot, by itself, establish that operator's physical truth.

5.5 Uncertainty and response

Conditional measurement uncertainty. Variance, inverse variance, or covariance of the admitted measurement conditional on fixed factor, atmosphere, identity, selection, and response state. Its propagation is exact for the fixed affine operator.

Nuisance uncertainty. Uncertainty in calibration, atmosphere, or response state, with named correlation scope. It is not included in a conditional weight unless explicitly incorporated through a documented joint covariance. Missing nuisance uncertainty is unavailable, not zero.

Response basis. The originating Beammap APT beam/template occurrence, including elliptical parameters and their units/frame when available, together with the separately supplied realized mapmaker, kernel, and filter response required to interpret a downstream point-source peak. A circularization is a labeled approximation, never silent replacement.

5.6 Validity and quality tuple

Every coherent observation or explicitly declared segment has a state tuple

$$\mathcal{V} = (R, E, S, Q, U_c, U_n, B), \tag{1}$$

where R is requested mode, E effective mode, S calibrated-signal validity, Q one segment-level opacity quality class, U_c conditional uncertainty availability, U_n nuisance/total-uncertainty completeness, and B response-basis completeness. Required reason codes accompany every non-success state. In particular:

- **science-qualification-eligible** means structurally admissible under the v0.1 policy with fully resolved scientific inputs. It is an eligibility state, not an achieved `science-qualified` or `calibrated-science` claim. Unless the owner later defines a different explicit threshold, either achieved claim additionally requires separately accepted atmosphere-representation-fidelity and observational-performance evidence over the same declared identity and support;
- **engineering-only unavailable** means the opacity is in $0.15 < \tau_{225} \leq 0.25$, for which no calibrated SCI-CAL output is authorized in v0.1;

- **outside-supported calibration** means the request lies outside an admitted domain or lacks a required operator/input identity;
- **disabled/uncalibrated** records an intentional effective no-op and never relabels uncalibrated numbers; and
- **uncertainty unavailable** may coexist with a valid calibrated signal when the missing item is not required to define the signal, but it prohibits the corresponding uncertainty or significance claim.

5.7 Three claim layers

Table 3: Claims that must not be conflated.

Claim	What it establishes	What it does not establish
Structural calibration correctness	typed inputs, valid association, exact factor algebra, once-only lineage, supported operator evaluation, uncertainty transform, and reconstructible output	truth of the atmosphere model, telescope response, repeatability, absolute flux accuracy, or production readiness
Atmosphere representation fidelity	agreement of the declared finite operator with a named higher-fidelity atmosphere calculation over declared support	agreement with the actual atmosphere, WVR accuracy, telescope passband truth, or source recovery
Observational calibration performance	repeatability and absolute recovery against independent astronomical evidence with full covariance	structural correctness by itself or universal performance outside the studied population and support

The prior goals of at most about one-percent atmosphere representation error, about five-percent relative repeatability, and about five-to-ten-percent absolute accuracy per band are validation targets only. They are not guarantees and are not used to fill missing evidence.

6 Assumptions, conditions, and unavailable authority

The equations are conditional on the following explicit statements. They are not permissions to replace missing information with defaults.

`SCI-CAL-ASM-001`: **upstream sample authority.**
: The supplied sample axis, time, eligibility, original/synthesized state, elevation/airmass meaning, and acquisition-column binding are correct under approved upstream authority. SCI-CAL neither chooses nor rederives them.

`SCI-CAL-ASM-002`: **ordinary measured channel.**
: The admitted signal is the ordinary `xs` stream. A distinct measured stream is a distinct physical observable and does not inherit this factor.

`SCI-CAL-ASM-003`: **selected immutable APT.**
: TolProj has selected and bound one immutable measured APT to the target observation, and the supplied target-to-source associations are admitted TolAPT/TolProj results. SCI-CAL does not choose the artifact or matcher.

`SCI-CAL-ASM-004`: **explicit affine convention.**
: Any baseline or offset operation preceding the multiplier is named, typed, and already authorized. "No baseline subtraction" is an explicit not-applicable state, not an inferred zero.

`SCI-CAL-ASM-005`: **coefficient direction and reference plane.**
: The selected `flxscale` is finite, nonzero, oriented as $\mathrm{mJy\,beam}^{-1}/U_x$, and refers to the top-of-atmosphere calibration plane. Its lineage states the disposition of any TolProj pointing-derived correction.

`SCI-CAL-ASM-006`: **atmosphere inputs.**
: τ_{225} is zenith optical depth; X is the full eligible-sample airmass; the target pivot is $X_{\mathrm{ref}} = 0$; and every value

carries declared time and model support. SCI-CAL does not infer X from elevation or choose an opacity estimator.

`SCI-CAL-ASM-007`: fixed conditional state.
Conditional covariance equations hold at fixed identities, factor values, atmosphere state, selection, and response state. Dependence of fitted state on the same signal requires cross-covariance or joint propagation.

`SCI-CAL-ASM-008`: response ownership.
Beam/Beammap supplies the originating beam/template identity; MAP/FLT supplies the realized map-maker/kernel/filter response. SCI-CAL records and conditions on those identities but does not certify their empirical fidelity.

`SCI-CAL-ASM-009`: coherent segment.
A segment is declared before quality classification and is internally coherent in selected APT, atmosphere/operator identity, requested/effective mode, target unit, and response basis. Splitting a segment is a provenance event, not a sample-by-sample quality switch.

`SCI-CAL-ASM-010`: passband limitation.
The selected TolTECA v1 passband set is a modeled array-named reference with the recorded unknowns in Section 5. Any atmosphere result using it is conditional on the declared, presently unestablished response weighting and carries unavailable passband uncertainty.

`SCI-CAL-ASM-011`: unresolved numeric operator authority.
The packet does not contain the exact nodes, node ordinate orientation, content digest, or numeric line-of-sight support of `am12_fixed_djf25_piecewise_linear_los_tau_v1`. Consequently this draft defines the strongest structural operator contract but does not authorize a numeric atmosphere evaluation, calibrated numeric output, or numeric `science-qualification-eligible` disposition until the owner supplies one content-bound operator record answering the question in the draft decision register. Resolving that question can enable structural evaluation only; it does not establish an achieved `science-qualified` or `calibrated-science` claim.

7 Canonical equations and invariants

7.1 Admission and valid support

Let A_j be the conjunction of the scientific admission predicates: the sample is upstream-eligible `xs`; its acquisition key and sample column binding are resolved; its selected APT row and unique admitted association are resolved; units and factor directions are declared; the sample, baseline state, factor, opacity, airmass, time support, operator, and response basis are valid for the requested operation; and the effective mode is enabled. The mathematical domain is

$$\mathcal{J}_{\mathrm{valid}} = \{j : \mathsf{A}_j = \mathrm{true}\}. \tag{2}$$

Restriction to $\mathcal{J}_{\mathrm{valid}}$ is not multiplication by a Boolean mask. Invalid occurrences do not become scientific zeros and do not contribute hits or coverage.

The explicitly declared affine input is

$$z_j = x_j^{xs} - b_j, \qquad j \in \mathcal{J}_{\mathrm{valid}}. \tag{3}$$

If the upstream convention has no subtraction, its record says that the baseline operation is not applicable and Equation (3) reduces explicitly to $z_j = x_j^{xs}$. Missing baseline meaning is not that state.

7.2 Retained structural atmosphere operator

The line-of-sight coordinate at the top-of-atmosphere pivot is

$$\ell_{ot} = \tau_{225,o}(t)\,[X_{ot} - X_{\mathrm{ref}}] = \tau_{225,o}(t) X_{ot}, \qquad X_{\mathrm{ref}} = 0. \tag{4}$$

This is a coordinate for the retained operator, not a claim that a TolTEC band opacity equals τ_{225}, nor permission to replace the retained operator by an unapproved monochromatic exponential.

For each array, a complete content-bound record for $\mathcal{O}_a$ must declare strictly ordered nodes

$$0 = \ell_{a,0} < \ell_{a,1} < \cdots < \ell_{a,K_a}, \qquad H_{a,0} = 1, \tag{5}$$

their dimensionless ordinates $H_{a,k}$, and one orientation $\sigma_a \in \{\text{transmission}, \text{correction}\}$. On a closed node interval,

$$u = \frac{\ell - \ell_{a,k}}{\ell_{a,k+1} - \ell_{a,k}}, \tag{6}$$

$$H_a(\ell) = (1-u)H_{a,k} + uH_{a,k+1}, \quad \ell \in [\ell_{a,k}, \ell_{a,k+1}]. \tag{7}$$

The canonical transmission and correction are then

$$(T_a, C_a) = \begin{cases} (H_a, H_a^{-1}), & \sigma_a = \text{transmission}, \\ (H_a^{-1}, H_a), & \sigma_a = \text{correction}. \end{cases} \tag{8}$$

Interpolation is applied to the authoritative ordinate before taking a reciprocal. Interpolating the reciprocal instead is, in general, a different operator and is not conformant.

The structural invariants are

$$T_a(0) = C_a(0) = 1, \qquad 0 < T_a(\ell) < 1, \quad 1 < C_a(\ell) < \infty \quad \text{for } 0 < \ell \le \ell_{a,K_a}, \tag{9}$$

with T_a strictly decreasing and C_a strictly increasing on declared support. Thus there is no positive-opacity unity plateau. Exact node values are reproduced and the operator is continuous at every seam. Its derivative may change discontinuously at a seam.

The authorized structural domain for a sample is the intersection

$$\mathcal{D}_{otd} = \left\{ 0 \le \tau_{225,o}(t) \le 0.15, \quad X_{ot} \in \mathcal{D}_X, \quad 0 \le \ell_{ot} \le \ell_{a(d),K_{a(d)}}, \quad t \in \mathcal{D}_\tau \right\}, \tag{10}$$

where $\mathcal{D}_X$ is the declared upstream airmass-model domain and $\mathcal{D}_\tau$ is the opacity source's declared time support. A time-resolved opacity at a sample is admissible only when measured at that time or obtained by a source-authorized interpolation between bracketing states whose declared validity covers the sample. No extrapolation from an unbracketed endpoint is allowed. Equation (10) is exact as a support intersection, but cannot be numerically instantiated until the operator record in `SCI-CAL-ASM-011` is supplied.

7.3 Once-only signal factorization

The selected APT is the value boundary for absolute calibration. If TolProj has materialized a pointing-derived correction, its causal ancestry may be written

$$F_{od}^{\text{sel}} = P_{od} F_{od}^{\text{parent}} \quad \text{(lineage identity only)}. \tag{11}$$

The right-hand factors are not separately applied to the target signal. Admissible correction dispositions are `not_applicable` and `embodied_once`, with the parent and event identities present for the latter. `required_but_unresolved` is not an admissible calibrated state.

Because v0.1 has only the target unit mJy beam^{-1}, its unit-transfer factor is the explicit identity $Q_{\text{mJy beam}^{-1}} = 1$. The exact realized multiplier and signal are

$$M_j = F_{od}^{\text{sel}}\, C_{a(d)}(\ell_{ot})\, Q_{\text{mJy beam}^{-1}}, \tag{12}$$

$$y_j = M_j z_j, \qquad j \in \mathcal{J}_{\text{valid}}. \tag{13}$$

No independent `responsivity`, `sens`, parent `flxscale`, pointing-derived correction, or opaque `fcf` appears in Equation (12). Any such quantity used upstream may be named in lineage or a nuisance model, but does not become another signal factor.

Numeric multiplication in Equation (12) is commutative. Causal provenance is not: the selected APT must exist before its row factor is bound, and target atmosphere state is applied at the target sample. For each factor role $\rho \in \{\text{absolute flxscale}, \text{target atmosphere}, \text{unit identity}\}$, define the realized application count $n_{j\rho}$. The once-only invariant is

$$n_{j\rho} = 1 \quad \text{for every required role and } j \in \mathcal{J}_{\text{valid}}, \tag{14}$$

and zero for roles not in the canonical multiplier. The factor-instance identities, not merely their numeric values, establish this count.

For a nonzero intended factor f, challenge results relative to the correct output obey

$$\frac{y_{\text{omitted}}}{y_{\text{correct}}} = f^{-1}, \qquad \frac{y_{\text{duplicated}}}{y_{\text{correct}}} = f, \qquad \frac{y_{\text{inverted}}}{y_{\text{correct}}} = f^{-2}, \tag{15}$$

when all other quantities are fixed and the inverse replaces the intended factor. A unity factor leaves values unchanged but must still retain its role and instance identity when the role is required.

7.4 Conditional measurement uncertainty

Stack the valid samples and let $A = \operatorname{diag}(M_j)$. Conditioned on fixed state θ,

$$\mathbb{E}[\boldsymbol{y} \mid \theta] = A\{\mathbb{E}[\boldsymbol{x}^{xs} \mid \theta] - \boldsymbol{b}\}, \tag{16}$$

$$C_{y|\theta} = AC_{x|\theta}A^{\mathsf{T}}. \tag{17}$$

This transformation preserves every supplied sample, detector, and observation correlation. For a diagonal conditional representation,

$$v_j^{(y)} = M_j^2 v_j^{(x)}, \qquad w_j^{(y)} = \frac{w_j^{(x)}}{M_j^2}. \tag{18}$$

The valid support is identical to the signal support. If the corresponding input uncertainty is absent, the output state is unavailable rather than a zero variance or infinite weight.

Every synthetic injection, conditional noise realization, or genuine raw-domain Jacobian companion $k_{jr} = \partial x_j / \partial \alpha_r$ that is declared to share the signal response obeys

$$k_{jr}^{(y)} = M_j k_{jr}^{(x)}. \tag{19}$$

This equation does not authorize relabeling a physically distinct measured stream as a response companion.

7.5 Calibration and response nuisance covariance

For a fixed factor sign, let

$$H_{jr} = \frac{\partial \ln |M_j|}{\partial \theta_r}. \tag{20}$$

Within one atmosphere interpolation interval define

$$\gamma_a(\ell) = \frac{d \ln C_a}{d\ell} = \begin{cases} -H_a'(\ell)/H_a(\ell), & \sigma_a = \text{transmission}, \\ H_a'(\ell)/H_a(\ell), & \sigma_a = \text{correction}. \end{cases} \tag{21}$$

Hence, away from seams,

$$\frac{\partial \ln C_a}{\partial \tau_{225}} = \gamma_a X, \qquad \frac{\partial \ln C_a}{\partial X} = \gamma_a \tau_{225}. \tag{22}$$

For mean calibrated signal $\boldsymbol{\mu}$, the first-order nuisance term is

$$C_{\text{nuis}} \simeq \operatorname{diag}(\boldsymbol{\mu}) H C_\theta H^{\mathsf{T}} \operatorname{diag}(\boldsymbol{\mu}). \tag{23}$$

It is generally dense. A common fractional absolute-scale uncertainty s gives

$$C_{\text{common}} = s^2 \boldsymbol{\mu}\boldsymbol{\mu}^{\mathsf{T}}, \tag{24}$$

a rank-one contribution that does not decrease merely because more affected samples are averaged.

If fitted calibration state or baseline depends on the same data, a first- order total covariance also contains cross terms:

$$C_y \simeq AC_xA^{\mathsf{T}} + J_\theta C_\theta J_\theta^{\mathsf{T}} + J_b C_b J_b^{\mathsf{T}} + AC_{x\theta}J_\theta^{\mathsf{T}} + J_\theta C_{\theta x} A^{\mathsf{T}} + \text{other declared cross terms}, \tag{25}$$

where $J_b = -A$. Independence can remove a cross term only when justified. Linearization is unavailable at an operator seam, for uncertainty spanning multiple intervals, for large or asymmetric multiplicative uncertainty, or for discrete association and selection uncertainty. Those cases require nonlinear samples, a joint posterior, or a bounded sensitivity study.

7.6 Response condition for the unit label

Let $P_B(\boldsymbol{\omega})$ be the originating Beammap response template, with declared coordinates and unit peak,

$$P_B(\mathbf{0}) = 1. \tag{26}$$

For an unresolved source of top-of-atmosphere flux density S mJy, a downstream realized linear response L has ideal map peak

$$m_{\text{peak}} = S\rho_L \text{ mJy beam}^{-1}, \qquad \rho_L = [LP_B](\mathbf{0}). \tag{27}$$

The original point-source-peak label remains scientifically literal only when $\rho_L = 1$ on the declared support. If a kernel or filter changes ρ_L, the consumer must either propagate an explicit response renormalization and update the response-basis provenance, or withhold the literal peak-calibration claim. SCI-CAL does not estimate ρ_L.

For an explicitly selected elliptical Gaussian approximation, the originating solid angle would be

$$\Omega_B = \frac{\pi\theta_{\text{maj}}\theta_{\text{min}}}{4\ln 2}, \tag{28}$$

with angle units, frame, position angle, and approximation label retained. Equation (28) records response basis only. It does not authorize a surface-brightness conversion or integrated photometry in v0.1.

7.7 Canonical lineage reconstructibility

Let $\mathcal{L}_o$ denote one canonical calibration-lineage record per coherent reduction package. For every valid occurrence, it must be possible to resolve

$$\mathcal{L}_o \longrightarrow \{x_j^{xs}, b_j, K^{\text{acq}}, R^{\text{APT}}, E^{\text{assoc}}, F^{\text{sel}}, \mathcal{O}_a, \tau_{225}, X, C_a, Q, \mathcal{V}, \theta, P_B, L\} \longrightarrow M_j \longrightarrow y_j. \tag{29}$$

The arrows mean resolvable scientific lineage, not a required file layout. Compact product links may point to the canonical record and immutable APT; they need not duplicate dense tables in every product.

8 Normative scientific requirements

The words “must”, “must not”, “required”, and “invalid” are normative. These requirements define scientific conformance without prescribing a file, class, function, language, or storage layout.

8.1 Scope, state, and input typing

`SCI-CAL-REQ-001`: measured-channel scope.
The v0.1 operation must accept only the ordinary `xs` measured stream. A physically distinct measured stream must be rejected as unsupported and must not inherit the `xs` factor, unit, response, or uncertainty label.

`SCI-CAL-REQ-002`: target-unit scope.
The only calibrated target is top-of-atmosphere point-source-peak mJy beam^{-1} at $X_{\text{ref}} = 0$. An unsupported unit or photometric meaning must yield no calibrated SCI-CAL output, not a relabeled numeric array.

`SCI-CAL-REQ-003`: complete signal declaration.
Each admitted signal must declare its acquisition identity, pre-CAL unit, sample and detector shape, time/support, upstream eligibility, and any authorized offset/baseline convention. Missing or incompatible declarations invalidate the affected scope.

`SCI-CAL-REQ-004`: explicit semantic states.
Missing, disabled, automatic, unavailable, invalid, engineering-only, and science-qualification-eligible states must be represented explicitly. None may be encoded solely as a zero, unity, negative, non-finite, stale, or undocumented sentinel.

`SCI-CAL-REQ-005`: one-way resolution.
Requested, effective, observation-resolved, and realized calibration states must remain distinct and flow one way. Realized or compatibility state must not mutate the accepted request, and a new observation must replace its resolved state in full.

8.2 Occurrence-scoped identity and association

`SCI-CAL-REQ-006`: acquisition key.
Every selected detector occurrence must have an observation-local key containing observation/Tune context, network or interface, and the network-local channel/tone slot. A global column is only a locator.

`SCI-CAL-REQ-007`: binding evidence.
A sample column may be bound to an acquisition occurrence and selected APT row only by an explicit keyed relation or a verified ordered-row contract that records scope, order definition, cardinality, and proof. Row position without that evidence is invalid.

`SCI-CAL-REQ-008`: unique coverage.
On the declared selected support, binding and target-to-source association must be single-valued and complete. Missing, duplicate, ambiguous, or contradictory keys must invalidate exactly the affected detector or the predeclared required support; no arbitrary choice is permitted.

`SCI-CAL-REQ-009`: immutable selected APT identity.
The selected APT must be identified as one immutable artifact by resolvable artifact identity and digest. Each used row must also carry its artifact- local key and source observation/network occurrence context.

`SCI-CAL-REQ-010`: association evidence.
Each admitted target-to-source edge must name both occurrence-scoped endpoints, matcher method/version, disposition, and available quality evidence. Only an admitted unique match may calibrate; abstained, ambiguous, rejected, or missing edges must not.

`SCI-CAL-REQ-011`: layered identity.
Acquisition, measured-APT, association, and design identities, as well as occurrence, semantic-content, and byte-transport identities, must remain distinct. A design match is not required for ordinary measured Beammap calibration, and a design-ID change alone must not change a valid result.

`SCI-CAL-REQ-012`: array/network consistency and lifetime.
Network-to-array membership and target/source network context must be checked as declared constraints. Reordering is harmless only when the binding proof remains valid. No identity or row state may leak from another observation.

8.3 Factor authority, causality, and reconstructibility

`SCI-CAL-REQ-013`: absolute factor.
The sole absolute detector factor must be the selected APT row's finite, nonzero `flxscale`, with direction mJy beam$^{-1}/U_x$, detector recipient, validity, reference plane, uncertainty status, and selected-artifact lineage declared. Reciprocal or sign convention must never be inferred.

`SCI-CAL-REQ-014`: pointing-correction disposition.
The selected lineage must classify any TolProj pointing-derived correction as `not_applicable`, `embodied_once`, or `required_but_unresolved`. An embodied correction must name its parent and event and must not be applied again; an unresolved required correction prohibits calibrated output.

`SCI-CAL-REQ-015`: once-only application.
Absolute `flxscale`, target atmosphere correction, and the explicit target-unit identity must each have realized application count one on every valid occurrence. Parent factors and already embodied events must have runtime count zero.

`SCI-CAL-REQ-016`: canonical multiplier.
The signal must obey Equations (12) and (13) on exactly $\mathcal{J}_{\mathrm{valid}}$. Scalar multiplication order may change without changing values, but the causal lineage and factor-instance identities must be preserved.

`SCI-CAL-REQ-017`: responsivity boundary.
`responsivity` must not be presented or applied as an independent absolute mJy beam^{-1} factor. Any use must have a distinct relative role, unit, normalization/recipient, support, and lineage, and must not duplicate information already embodied in `flxscale`.

`SCI-CAL-REQ-018`: sensitivity boundary.
Beammap `sens` must not multiply the calibrated signal or masquerade as atmosphere correction or total uncertainty. If used for an approximate weight, that role, units, estimator identity, approximation, support, and exclusions must be explicit.

`SCI-CAL-REQ-019`: compatibility factor boundary.
An `fcf` or other compatibility aggregate may be admitted only after exact decomposition into the canonical

role, unit, direction, constituent instance identities, and exclusions. An opaque aggregate has no calibration authority.

`SCI-CAL-REQ-020`: factor and package lineage.
Every realized factor must record definition, role, unit, direction, recipient/support, reference plane, application stage, validity, uncertainty status, and parent lineage. One canonical record per coherent reduction package must reconstruct Equation (29); product-level links may resolve it without duplicating the complete APT.

8.4 Atmosphere operator, support, and quality

`SCI-CAL-REQ-021`: opacity and airmass meaning.
τ_{225} must be finite nonnegative zenith optical depth with source and time support. X must be the full eligible-sample airmass with model identity/domain. Equation (4) and $X_{\mathrm{ref}} = 0$ must be used; a zenith value alone is not a line-of-sight correction.

`SCI-CAL-REQ-022`: temporal support.
Every calibrated sample must be covered by an exact opacity state or a source-authorized interpolation between valid bracketing states. Method, version, endpoints, validity interval, and any maximum gap must be recorded. Unbracketed interpolation, endpoint extrapolation, and inherited prior- observation opacity are prohibited.

`SCI-CAL-REQ-023`: complete operator identity.
A numeric atmosphere evaluation requires the exact operator name, content digest, array assignment, ordered coordinate/ordinate nodes with units, ordinate orientation, interpolation rule, numeric support, generating model provenance, and passband identity when used. A name without these facts is insufficient.

`SCI-CAL-REQ-024`: structural evaluation.
The retained operator must obey Equations (5) through (8): analytic zero anchor, interpolation of the authoritative ordinate, exact node recovery, and continuity at seams.

`SCI-CAL-REQ-025`: physical-direction invariants.
On declared positive-opacity support, transmission and correction must be finite and positive, transmission must be strictly below unity and decreasing, and correction must be strictly above unity and increasing. A finite unity plateau, sign change, non-finite value, or direction reversal is invalid.

`SCI-CAL-REQ-026`: domain intersection.
A calibrated sample must satisfy Equation (10). No opacity, elevation/airmass, line-of-sight coordinate, or time request may be silently clamped, extrapolated, or replaced by a legacy selector/fallback.

`SCI-CAL-REQ-027`: science-qualification eligibility policy.
A coherent segment may receive the `science-qualification-eligible` structural-policy class only when all its admitted opacity support lies in $0 \le \tau_{225} \le 0.15$ and every other structural admission condition holds. This class records eligibility to seek separately accepted qualification evidence; it is not an atmosphere-fidelity, observational-performance, `science-qualified`, or `calibrated-science` claim.

`SCI-CAL-REQ-028`
Engineering-opacity restriction. If any part of an unsplit coherent segment has $0.15 < \tau_{225} \le 0.25$, its class must be `engineering-only unavailable` and v0.1 must emit no calibrated SCI-CAL signal. It must not extrapolate, fall back, or preserve a calibrated unit label.

`SCI-CAL-REQ-029`: invalid and outside-supported opacity.
Negative or non-finite opacity is invalid input. Opacity above 0.25, absent opacity, invalid airmass/elevation meaning, or a request beyond any operator node/time/model support is outside supported calibration and must yield no calibrated output.

`SCI-CAL-REQ-030`: coherent quality state.
One predeclared observation or segment must receive one opacity quality class. It must take the most restrictive applicable state across its support; sample-by-sample class switching is prohibited. A separately declared split creates separate lineage-bearing segments.

`SCI-CAL-REQ-031`: passband limits.
Where the operator depends on a passband, it must bind the exact selected TolTECA v1 passband-set identity and record all known unknowns in Section 5. No detector/network/telescope measurement, normalization, uncertainty, covariance, or photon/energy semantics may be inferred from the array-named curves.

8.5 Conditional and nuisance uncertainty

`SCI-CAL-REQ-032`: full conditional covariance.
A supplied conditional covariance must transform by Equation (17) on the identical ordered valid support. Existing off-diagonal correlations must be preserved.

`SCI-CAL-REQ-033`: scalar variance and weight.
Supplied diagonal conditional variance or inverse variance must transform by Equation (18), with output units $(\mathrm{mJy\,beam}^{-1})^2$ or $(\mathrm{mJy\,beam}^{-1})^{-2}$. Weight is not support, validity, hits, response, confidence, or total uncertainty.

`SCI-CAL-REQ-034`: unavailable conditional uncertainty.
If no admissible variance, weight, or covariance exists for a valid signal, conditional uncertainty must be marked unavailable on that same support. It must not be synthesized as zero variance, infinite weight, or a value derived from `sens` without a separately approved estimator.

`SCI-CAL-REQ-035`: nuisance completeness ledger.
The lineage must carry an entry for detector `flxscale`, common absolute calibrator scale, any TolProj pointing-derived correction, WVR/atmosphere model, Beammap `sens` when used for approximate weights, and beam/template response. Each entry must be `quantified`, `not_applicable` with justification, or `unavailable`, and name detector, array, observation, cohort, or global correlation scope when known.

`SCI-CAL-REQ-036`: correlation preservation.
Nuisance covariance must preserve declared common and cross terms. Shared calibrator, array, opacity, pointing, passband, or beam terms must not be treated as independent per sample or counted down solely with sample or detector number.

`SCI-CAL-REQ-037`: total-uncertainty claim.
Total calibrated uncertainty or significance is available only when every required nuisance category and material cross-covariance is quantified or explicitly shown not applicable. Otherwise it must be labeled unavailable, even when conditional weight exists.

`SCI-CAL-REQ-038`: propagation regime.
Linear nuisance propagation may be claimed only inside one differentiable operator interval with sufficiently small approximately symmetric uncertainties and justified dependence assumptions. Seam-spanning, large, asymmetric, selection, association, or same-data fitted-state uncertainty requires nonlinear or joint propagation; missing cross terms may not be set to zero without justification.

`SCI-CAL-REQ-039`: shared realized operator.
Signal, conditional covariance/weights, admitted synthetic injections, conditional noise realizations, and genuine Jacobian companions must use the identical realized multiplier, valid support, and factor lineage. A different measured stream must not be substituted as a companion.

8.6 Response basis, validity, and claims

`SCI-CAL-REQ-040`: originating response basis.
Every calibrated result must resolve its selected Beammap source and per- detector beam/template occurrence. Available elliptical major/minor axes, position angle, units, coordinate frame, epoch/support, fit identity, and uncertainty status must be retained.

`SCI-CAL-REQ-041`: realized response distinction.
Originating Beammap response and realized mapmaker/kernel/filter response must remain separate provenance objects. Circularization must be explicitly labeled with its rule and cannot silently replace available elliptical information.

`SCI-CAL-REQ-042`: point-source-peak condition.
A downstream result may claim literal point-source-peak $\mathrm{mJy\,beam}^{-1}$ only when its realized response satisfies Equation (27) with $\rho_L = 1$ for the declared beam/template and support, or when an explicit response renormalization establishes and records that condition.

`SCI-CAL-REQ-043`: response-changing operations.
A kernel, filter, or map operation that changes peak response must propagate the response basis and factor. Without a separately established renormalization it must withhold the literal peak-calibration claim; SCI-CAL itself must not certify the downstream response.

`SCI-CAL-REQ-044`: excluded photometric meanings.
V0.1 must not emit or authorize MJy/sr, Jy/pixel, Rayleigh-Jeans or thermodynamic temperature, extended-

source calibration, integrated photometry, or a resolved-source recovery claim.

`SCI-CAL-REQ-045`: validity tuple and causes.
Every coherent result must publish the tuple in Equation (1) and machine-distinguishable reason codes for missing identity, unsupported stream/unit, invalid or out-of-domain atmosphere, intentionally disabled calibration, unavailable conditional or nuisance uncertainty, incomplete response basis, and any required output failure.

`SCI-CAL-REQ-046`: truthful no-output states.
Disabled/uncalibrated, invalid, engineering-only unavailable, outside- supported calibration, and uncertainty-unavailable states must remain distinguishable. A failure in a required factor or identity must not emit a calibrated number; a missing non-signal uncertainty may leave the signal valid only with the corresponding claim explicitly unavailable.

`SCI-CAL-REQ-047`: canonical lineage record.
One resolvable canonical record per coherent reduction package must include raw observation identity, selected APT and digest, acquisition and selected row occurrences, association edge, factor definitions and ancestry, atmosphere/operator/passband identities and support, reference plane, target unit, state tuple, uncertainty ledger, and response basis.

`SCI-CAL-REQ-048`: compact product links.
Every calibrated product must contain links and occurrence keys sufficient to resolve its canonical lineage and exact selected APT. It need not and should not duplicate the entire APT or dense per-detector lineage table in every product.

`SCI-CAL-REQ-049`: separated claim status.
Structural correctness, atmosphere representation fidelity, relative repeatability, and absolute flux performance must be reported as separate claims with separate evidence and support. A pass in one layer must not be promoted to another. In particular, `science-qualification-eligible` must not be promoted to an achieved `science-qualified` or `calibrated-science` claim. Unless the owner later defines a different explicit threshold, either achieved claim requires separately accepted atmosphere-representation-fidelity and observational-performance evidence over the same declared identity and support in addition to structural correctness.

`SCI-CAL-REQ-050`: falsifiability before qualification.
Any conformance or performance claim must identify evidence capable of falsifying the relevant equations, assumptions, edge cases, identity invariants, uncertainty correlations, and response condition. Skipped required evidence is unavailable, not a pass; the provisional one-percent, five-percent, and five-to-ten-percent targets are not guarantees.

9 Limiting cases, pathologies, and falsifiable predictions

The following predictions are consequences of the common authority. They are pre-validation statements, not reports of executed evidence. “No output” means no value carrying a calibrated SCI-CAL unit label; a diagnostic or explicitly uncalibrated passthrough may exist only as a different product state.

Table 4: Required limiting and pathological outcomes.

ID	Challenge	Falsifiable prediction
`SCI-CAL-EDGE-001`	Zero opacity	For every supported array and finite admitted airmass, $\ell = 0$, $T = C = 1$, and atmosphere correction changes neither signal nor conditional uncertainty. With every other structural condition satisfied, this may establish only `science-qualification-eligible`; it does not establish an achieved `science-qualified` or `calibrated-science` claim without the separately accepted evidence in `SCI-CAL-REQ-049`.
`SCI-CAL-EDGE-002`	Increasing opacity	At fixed admitted $X > 0$, increasing positive τ_{225} strictly decreases T and increases C while remaining finite. Any finite positive-opacity unity plateau is a failure.

ID	Challenge	Falsifiable prediction
SCI-CAL-EDGE-003	Zenith and airmass	At zenith $X = 1$, $\ell = \tau_{225}$. At fixed positive opacity, a larger admitted airmass gives larger ℓ and larger correction. At zero opacity the result is independent of airmass.
SCI-CAL-EDGE-004	Exact atmosphere node	Evaluation at $\ell_{a,k}$ returns the declared authoritative ordinate exactly, followed by the one declared orientation mapping. Interpolation of the reciprocal instead is detectable at nontrivial intervals.
SCI-CAL-EDGE-005	Operator seam	Left and right values agree at every node. Slopes may differ; a derivative or linear covariance claim exactly at, or spanning, a seam is unavailable and requires nonlinear propagation.
SCI-CAL-EDGE-006	Invalid operator domain	A value outside numeric node support, outside the upstream airmass model, or outside declared time support yields no calibrated output. Clamping, extrapolation, and a legacy fallback are detectable failures.
SCI-CAL-EDGE-007	Bracketed opacity series	A sample between two valid source states is admissible only if the named source interpolation rule authorizes that bracket and its gap. Its lineage resolves both endpoints, method, and validity interval.
SCI-CAL-EDGE-008	Unbracketed opacity series	A sample before the first state, after the last state, across a disallowed gap, or covered by only one endpoint is outside temporal support. It yields no calibrated output rather than an endpoint hold or extrapolation.
SCI-CAL-EDGE-009	Engineering opacity	Any unsplit segment containing $0.15 < \tau_{225} \leq 0.25$ is `engineering-only unavailable`; v0.1 emits no calibrated signal for that segment.
SCI-CAL-EDGE-010	Outside or malformed opacity	Negative or non-finite opacity is invalid. Opacity above 0.25 or absent opacity is outside supported calibration. Neither case is silently clamped or relabeled.
SCI-CAL-EDGE-011	Unity factor	Replacing one required factor value by numeric unity leaves values unchanged only when unity is the intended value; the factor role and application count remain present and auditable.
SCI-CAL-EDGE-012	Omitted, duplicated, or inverted factor	For an isolated nonzero factor f, output ratios are exactly those in Equation (15). Signal and uncertainty discrepancies identify the challenged role.
SCI-CAL-EDGE-013	Corrected-APT double application	With $F^{\mathrm{sel}} = PF^{\mathrm{parent}}$, applying P again changes the result by P and violates application count. Using the parent without the embodied event changes it by P^{-1}.
SCI-CAL-EDGE-014	Keyed row permutation	Arbitrarily permuting selected APT rows while preserving row occurrence keys and the explicit keyed relation leaves every output, validity state, and nuisance recipient invariant.
SCI-CAL-EDGE-015	Ordered-row permutation	Permuting rows under an ordered-row binding without supplying a new proof invalidates the binding. A newly proven permutation may be admitted and then gives the keyed-equivalent result.
SCI-CAL-EDGE-016	Network or column reorder	Reordering network containers or sample columns while preserving acquisition keys and bindings leaves results invariant after restoring occurrence order. Inferring identity from the new position produces a detectable mismatch or failure.
SCI-CAL-EDGE-017	Missing or duplicate key	A missing acquisition/APT key, duplicate keyed candidate, inconsistent array/network constraint, or multiple sample columns for one selected occurrence invalidates the affected declared scope. No first-row or nearest-row choice is allowed.

ID	Challenge	Falsifiable prediction
SCI-CAL-EDGE-018	Association disposition	A unique admitted match may calibrate. Abstained, ambiguous, rejected, and missing associations produce no calibrated value for the endpoint; aggregate matcher performance cannot convert one into a match.
SCI-CAL-EDGE-019	Irrelevant design identity	Adding, removing, or changing a design-ID match while measured occurrence endpoints, association, selected APT values, and all other state remain fixed leaves the SCI-CAL result unchanged.
SCI-CAL-EDGE-020	Scalar conditional uncertainty	Multiplying signal by M multiplies variance by M^2 and divides inverse variance by M^2, including for a signed F^{sel}. Support and validity do not change.
SCI-CAL-EDGE-021	Vector conditional covariance	For a dense positive-definite input covariance, the output equals $AC_xA^\intercal$, including all off-diagonal signs and magnitudes. Replacing it with a diagonal is not the same claim.
SCI-CAL-EDGE-022	Unavailable uncertainty	Removing a conditional uncertainty input leaves it unavailable, not zero. Removing any required nuisance entry marks total calibrated uncertainty unavailable while a separately valid signal may remain.
SCI-CAL-EDGE-023	Common systematic and sample count	A shared fractional scale term gives Equation (24). Repeating or averaging identically affected samples does not reduce that fractional term as $N^{-1/2}$.
SCI-CAL-EDGE-024	Nonlinear nuisance case	Large/asymmetric factor uncertainty, uncertainty crossing an operator seam, or uncertain discrete association causes first-order propagation to be withheld. Posterior sampling or a bounded sensitivity calculation need not agree with a local linearization.
SCI-CAL-EDGE-025	Unresolved point source	Under a unit-peak originating beam and a realized response with $\rho_L = 1$, an unresolved source of flux density S mJy has ideal peak S mJy beam^{-1}. This is a response prediction, not an integrated-flux claim.
SCI-CAL-EDGE-026	Response-changing kernel	If a downstream kernel changes ρ_L to r, the unrenormalized ideal peak is rS mJy beam^{-1}. A literal peak claim requires recorded renormalization by r^{-1} and an updated response basis.
SCI-CAL-EDGE-027	Circularized beam	Replacing available elliptical axes by a circular beam without an explicit rule is nonconformant. With a declared circularization, the result is labeled approximate and does not erase the originating ellipse.
SCI-CAL-EDGE-028	Unsupported stream or target	A non-`xs` stream, or a request for surface brightness, per-pixel flux, temperature, extended-source calibration, or integrated photometry, yields an unsupported no-output state without a calibrated v0.1 label.
SCI-CAL-EDGE-029	Zero signal and affine convention	With an explicit not-applicable baseline and $x = 0$, every finite multiplier gives $y = 0$: CAL creates no additive offset. With a declared nonzero b, the result is $-Mb$, exposing the affine convention rather than hiding it.
SCI-CAL-EDGE-030	Observation-order replay	Two observations with distinct APT, association, atmosphere, and response identities give identical per- observation results in either processing order. No state or factor from one appears in the other's canonical lineage.

9.1 Proposed evidence, separated by claim

No scientific validation is executed by this authorship task. The following evidence is proposed for a future, separately authorized campaign.

V1. Analytic structural fixtures. Use small scalar and dense-vector cases to evaluate Equations (3), (7), (12), (17), and (18); include every edge in Table 4. Exact identities and metadata are compared exactly; numeric tolerances are preregistered for the arithmetic precision and never absorb a semantic mismatch.

V2. Metamorphic identity and lineage fixtures. Permute keyed rows, sample columns, and network containers; vary irrelevant design identity; challenge missing/duplicate keys and every association disposition; process observations in both orders; and inject omitted, duplicated, and inverted factor instances. Expected invariants and ratios are fixed by Equations (14) and (15).

V3. Atmosphere structural fixtures. Once the missing content-bound operator record is supplied, verify its digest, exact nodes, analytic zero anchor, positivity, strict direction, seam continuity, and support rejection without using observational data. This establishes structural correctness only.

V4. Atmosphere representation study. Compare the finite retained operator with its named higher-fidelity generating calculation on a dense, preregistered grid over the exact declared support, using the exact selected passband identity and a fully declared weighting convention. Report maximum and distributional error separately by array, opacity, and airmass. The prior one-percent target is a falsification threshold, not a guaranteed outcome. Unknown passband uncertainty remains a limitation even if the numeric approximation passes.

V5. Synthetic response and uncertainty study. Inject a point-source template through an independently declared beam and downstream response; test Equation (27) before and after a response-changing kernel. Inject detector, array, observation, and global nuisances and recover the dense covariance predicted by Equations (23) and (24). Blank inputs test the absence of additive offsets or residual airmass trend.

V6. Observational performance study. Only after structural and representation evidence exists, use independent standard-source predictions with epoch, spectrum, passband, beam/template, and full covariance. Estimate relative repeatability and absolute recovery separately by array and over declared atmosphere support; test residual trend with airmass. The prior approximate five-percent repeatability and five-to-ten-percent absolute goals are study targets, not acceptance by assertion. Correlated standards are not counted as independent.

10 Evidence package for a future audit or clean implementation

A review package can remain implementation neutral while exposing enough evidence to falsify the contract. The recommended evidence inventory is:

- immutable input and resolved-state records for at least two observations with deliberately different detector subsets, selected APTs, associations, atmosphere states, and response bases;
- the exact operator and passband content identities, node inventory, orientation, support, and opacity interpolation records;
- a machine-resolvable factor ledger showing role and application count, including one corrected-APT ancestry challenge;
- scalar and dense-vector conditional uncertainty fixtures, plus nuisance covariance fixtures at detector, array, observation, cohort, and global scopes;
- identity metamorphic cases covering keyed permutations, proven and broken ordered relations, network reorder, missing/duplicate keys, every association disposition, and irrelevant design-ID changes;
- response fixtures for an unresolved source before and after a known response-changing kernel; and
- output inventories, canonical lineage, compact links, validity tuples, reason codes, and claim-layer status for every case, including every truthful no-output state.

For each case, the evidence states expected equations and edge IDs before execution, numeric precision and tolerances, exact identity expectations, and whether a mismatch invalidates a detector, segment, or entire required output. A skipped required case remains unavailable. Structural fixtures do not stand in for the separately proposed atmosphere representation and observational studies in Section 9.1.

11 Conformance disposition

A future reviewer can use four non-overlapping dispositions:

Table 5: Disposition meanings.

Disposition	Meaning
Structurally conformant	All applicable numbered requirements and predeclared structural predictions pass on the exact reviewed candidate and support; this says nothing stronger about atmosphere fidelity or performance.
Nonconformant	One or more applicable equations, requirements, identity or support invariants, uncertainty transforms, lineage conditions, or truthful state outputs fail.
Unavailable	Required authority or evidence is absent. This is the current numeric-atmosphere disposition under `SCI-CAL-ASM-011`; it is not a pass or a failure of an untested implementation.
Not applicable	A requirement's premise is demonstrably absent and the canonical record gives the scientific justification, for example an embodied pointing correction marked `not_applicable`.

Atmosphere representation fidelity and observational performance receive separate dispositions and evidence. Their provisional numerical targets never convert "unavailable" into "pass".

12 Engineering conclusion

The engineering conformance surface is finite and falsifiable: typed input, layered identity, unique association, selected APT authority, once-only factor composition, supported atmosphere evaluation, identical uncertainty transfer, truthful quality/validity, response-basis preservation, and package-level reconstructibility. No current implementation mapping is part of this view.

Until the owner binds the missing operator record, the correct conformance behavior for any requested numeric atmosphere calibration is an explicit unavailable/no-calibrated-output state. Choosing nodes, orientation, support, or a fallback inside an implementation would be an unauthorized scientific decision. Binding that record can enable a structural disposition no stronger than `science-qualification-eligible`; it does not by itself establish an achieved `science-qualified` or `calibrated-science` claim.